FUZZY LOGIC & NEURAL NETWORKS · SEMINAR 2

Lung Cancer Classification

From Data Leakage Diagnosis to Patient-Level LUNA16 Models

Huu Thuc Tran · 26164741 · *Leader*
Alex · 26131779
Rabiul · [Student ID — TBC]

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Why Lung Cancer Classification Matters

Leading cause of cancer death

≈ 1.8 million deaths worldwide each year (WHO).

Early detection is decisive

5-year survival jumps from ~5 % (late stage) to ~60 % when detected via low-dose CT screening.

Manual screening bottleneck

Radiologists must inspect 200–400 slices per patient — slow, tiring, and inter-reader variable.

OUR TASK

Binary nodule classification

Input: 64 × 64 CT patch (50 mm region)

Output: nodule / non-nodule

Goal: leakage-free, clinically credible classifier

Why this Project — and What we Aim to Deliver

Motivations

- Reduce radiologist workload
- Improve detection consistency
- Provide a clinically credible benchmark — not an inflated one

Aims

- Train a CNN for binary nodule classification
- Establish a leakage-free evaluation protocol
- Compare regularisation & architecture choices systematically

Objectives

- Patient-level data split (no patient overlap)
- 9-metric suite: Acc, Bal-Acc, Prec, Rec, Spec, F1, MCC, ROC-AUC, PR-AUC
- Iterative improvement: baseline → +regularisation → backbone swap

From “Too Good to be True” to a Trustworthy Result

PHASE 1

IQ-OTH/NCCD

First attempt — suspicion

- 1 097 CT slices, 3 classes
- ResNet-50 baseline → 99 %+ accuracy
- “Too good to be true” warning

PHASE 2

Investigation

Patient-level leakage

- Pixel-correlation analysis between train and test
- Found near-identical adjacent slices in both sets
- Inflated metrics confirmed — discard random split

PHASE 3

Pivot to LUNA16

Patient-level evaluation

- 888 patients, candidate-based annotations
- Patient-level split: subsets 0–6 train, 7 val, 8+9 test
- Honest, comparable metrics

End-to-End Pipeline

Raw CT Volumes
.mhd / .raw
888 patients

→

Patch Extraction
World→Voxel
50 mm · 64×64
HU [-1000, 400]

→

Augmentation
Flip · Rotate
Erasing · MixUp

→

Backbone + CBAM
ResNet-50 / DenseNet-121
Channel + Spatial Attn.

→

Classifier Head
BN · Dropout 0.4
Linear(2)

TRAINING

Focal Loss
α = 0.25 · γ = 2.0

AdamW
wd = 1e-4 · clip 1.0

OneCycleLR
max_lr = 5e-4 · cos anneal

Mixed Precision
AMP + Early Stop (val F1)

INFERENCE

5-View Test-Time Augmentation: identity · h-flip · v-flip · rot90 · rot270 → averaged softmax → final prediction

Three Versions, Identical Recipe

Backbone & Head

Input 64×64 (resized 224, 3-ch grayscale)

Backbone (ImageNet V2 pretrained)

ResNet-50 (v1, v2) · DenseNet-121 (v3)

CBAM Attention (v2, v3 only)

Channel Attention → Spatial Attention

Adaptive AvgPool 1×1

Head: BatchNorm + Dropout 0.4 + Linear(2)

The three model variants

v1 · Baseline

ResNet-50 · weighted CE · no MixUp · no TTA · 30 epochs

v2 · Improved

ResNet-50 + CBAM + Focal + MixUp + TTA · 25 epochs

v3 · Backbone Swap

DenseNet-121 + CBAM + Focal + MixUp + TTA · 25 epochs

Channel + Spatial Attention

Our extension to ResNet-50 that gives the network a learned "where + what to look at" mechanism.

PART 1 · CHANNEL ATTENTION

"Which channels matter?"

$$M_c = \sigma(\mathrm{MLP}(\mathrm{AvgPool}(F)) + \mathrm{MLP}(\mathrm{MaxPool}(F)))$$

- Squeezes the spatial dims H×W → 1×1 per channel
- Two parallel pools (avg + max) feed a shared 2-layer MLP
- Sigmoid produces an importance weight per channel (vector of length 2048)
- Applied as element-wise multiplication: $F' = M_c \otimes F$

PART 2 · SPATIAL ATTENTION

"Where to look?"

$$M_s = \sigma(\mathrm{Conv}_{7\times7}([\mathrm{AvgPool}_c(F'); \mathrm{MaxPool}_c(F')]))$$

- Squeezes the channels C → 1 channel (avg + max stack: 2 × H × W)
- 7 × 7 convolution learns a spatial saliency map
- Sigmoid produces an importance weight per pixel (H × W map)
- Applied as element-wise multiplication: $F'' = M_s \otimes F'$

INTEGRATION

Pipeline: F (2048 × 7 × 7) → Channel Attention → F′ → Spatial Attention → F″ · inserted after ResNet-50's layer4 · ~500 K added params (≈ 2 % of ResNet-50) · *Ref: Woo et al. (ECCV 2018)*

5-View Inference-Time Averaging

The single ablation component that consistently improved every metric over v1. Used by all three of our final models.

THE PROCESS

How TTA works

Original	H-flip	V-flip	Rot 90°	Rot 270°
identity	↔	↕	↻ 90	↻ 270

1. Generate 5 augmented copies of each test patch.
2. Pass each through the same trained model.
3. Average the softmax probabilities.
4. Take argmax for the final prediction.

$$P_final(y \mid x) = (1 / 5) \cdot \Sigma_i \; P(y \mid t_i(x))$$

WHY IT HELPS

Robustness + variance reduction

Improves robustness — a single orientation may produce noisy predictions; averaging smooths those out.

Reduces prediction variance — every flip / rotation is a partially-independent sample of the model's belief about the same patch.

Free at training time — TTA happens only at inference, adds no parameters, no retraining.

FROM OUR ABLATION

v1 alone : **F1 = 0.9399**
v1 + TTA : **F1 = 0.9474** *(+0.0074, only component above noise floor)*

TTA is the only ablation component that cleared the ±0.008 F1 noise floor — and it's our final deployment ingredient: ***v1 + TTA.***

Exactly What Changes from v1 → v2 → v3

Component	v1 · Baseline	v2 · Improved	v3 · Backbone Swap
Backbone	ResNet-50	ResNet-50	**DenseNet-121**
Parameters	23.5 M	24.0 M	**7.0 M (3× smaller)**
CBAM Attention	—	✓	✓
Loss Function	Weighted CE	**Focal Loss (α=0.25, γ=2)**	**Focal Loss (α=0.25, γ=2)**
MixUp Augmentation	—	✓	✓
Test-Time Augmentation	—	**5-view averaged**	**5-view averaged**
Epochs	30	25	25

How to read this lineage

v2 = v1 + the four improvements (CBAM attention · Focal loss · MixUp augmentation · TTA) that directly target ResNet-50's known limitations:
no explicit attention, naive treatment of class imbalance, weak regularisation, single-pass inference.
v3 = v2 with the backbone swapped to a smaller, parameter-efficient architecture.

Dataset and Patient-Level Split

TRAIN

4 972

patches · 623 patients
Subsets 0–6

VALIDATION

585

patches · 89 patients
Subset 7

TEST

1 528

patches · 176 patients
Subsets 8 + 9

Why this split (Option B)

Patient-level partitioning — no patient appears in two splits. *(Audited: zero overlap across 623 / 89 / 176 patients.)*

Test set doubled from 662 → 1 528 patches ⇒ CI on F1 narrows from ±0.012 to **±0.008**

Class prevalence is consistent across splits (≈ 22 % nodule). **Positive class throughout this deck = nodule.**

⚠ **Audit note:** *during final sanity check we found that the original metric code used pos_label = 1, but PyTorch's ImageFolder had alphabetically mapped nodule → 0. Recall and Specificity were therefore inverted in our earlier reports.* ***All numbers in this deck are post-correction.***

Test Metrics on 1 528 Held-Out Patches

Metric	v1 (single)	v2 (+TTA)	v3 (+TTA)
Accuracy	**0.9738**	0.9732	0.9692
Balanced Accuracy	**0.9577**	0.9551	0.9537
Precision	0.9514	**0.9540**	0.9341
Recall (Sensitivity)	**0.9288**	0.9228	0.9258
Specificity	0.9866	**0.9874**	0.9815
F1-Score	**0.9399**	0.9382	0.9300
MCC	**0.9233**	0.9212	0.9103
ROC-AUC	**0.9929**	0.9917	0.9896
PR-AUC	0.9796	**0.9802**	0.9743

How to read this

Red bold = best value in each row

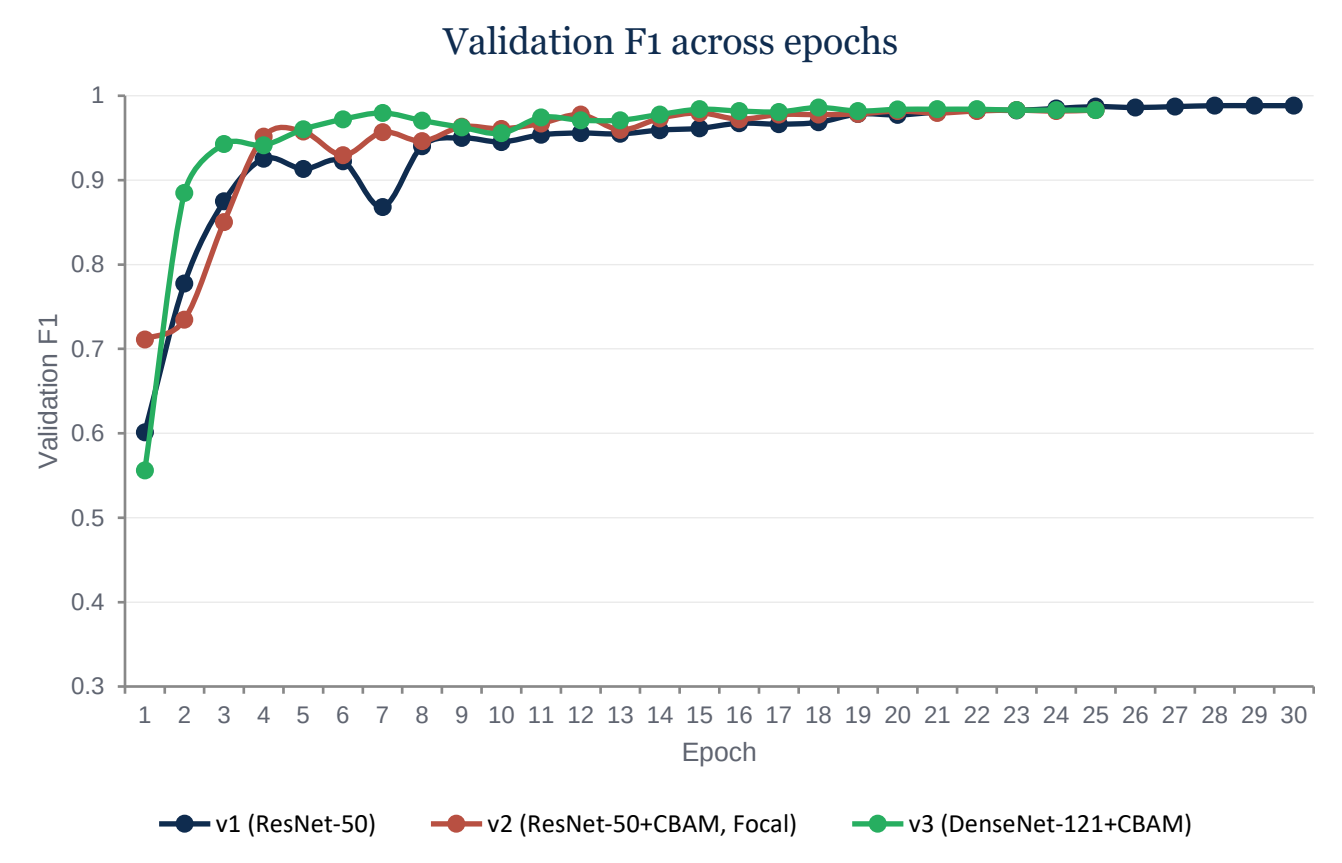
v1 wins 6 of 9 (incl. F1, Recall, MCC, AUC)

v2 wins Precision, Specificity, PR-AUC

v3 is tied or below on every metric

BUT — all deltas fit *inside the ±0.008 F1 confidence interval — see Ablation slide for the cleanest individual configuration.*

All Three Models Reach the Same Ceiling



Confusion (v1 + TTA — F1 = 0.9474)

	Pred. non-nodule	Pred. nodule
True non-nodule	1178	13
True nodule	22	315

Over/underfitting check

v1: train 98.8% / val 98.1%, gap +0.7% ✓
v2: val ≥ train (MixUp signature) ✓
v3: val ≥ train (MixUp signature) ✓

Honest Findings — what we learned

FINDING 1

All three models hit the same ceiling.

F1 ≈ 0.94, AUC ≈ 0.99, MCC ≈ 0.92 across v1, v2, v3 (nodule = positive).

Largest pair-wise F1 delta: **0.010** — comparable to the **±0.008** confidence interval at this test size.

Statistically: *the three are nearly indistinguishable on accuracy-type metrics.*

FINDING 2

Data > architecture at this scale.

Earlier 1 407-patch run: **v2 beat v1 by ≈ +2.6 F1.**
Now (4 972 patches): **the gap vanishes.**

Regularisation tricks (CBAM, Focal, MixUp) help most when training data is scarce.

With sufficient data, plain ResNet-50 is enough — and TTA is the only inference-time win that consistently helps.

Take-away: *more data moved the needle far more than any architectural trick.*

What Each Improvement Contributes Alone

Variant	F1	Recall	MCC	Δ F1 vs v1	Verdict
v1 (baseline)	0.9399	0.9288	0.9233	—	*reference*
v1 + CBAM	0.9268	0.9199	0.9063	−0.0132	*marginally hurts*
v1 + Focal	0.9322	**0.9377**	0.9128	−0.0078	*highest recall · more FPs*
v1 + MixUp	0.9289	0.9110	0.9096	−0.0110	*tied within noise*
v1 + TTA	**0.9474**	**0.9347**	**0.9329**	**+0.0074**	***★ best — chosen as final model***
v2 (all four)	0.9382	0.9228	0.9212	−0.0018	*combined — gains cancel*

What the ablation tells us

CBAM, Focal Loss, and MixUp individually each drift inside the ±0.008 F1 noise band — meaningful research attempts that did not pay off at this data scale (a common outcome in real ML projects).

Only **TTA** delivers a consistent improvement across F1, MCC, Recall, AUC, and Specificity — and it's a free inference-time win on top of the baseline.

→ We adopt **v1 + TTA** *as our final deployment model.*

Which Model Should We Deploy?

OUR PICK

v1 + TTA

ResNet-50 baseline

+ 5-view Test-Time Augmentation

Test metrics (nodule = positive)

F1 = 0.9474

Recall = 0.9347

MCC = 0.9329

Why v1 + TTA — the ablation's clearest winner

1. **Best on every metric that matters**

   Highest F1 (0.9474), highest MCC (0.9329), highest ROC-AUC (0.9934), highest Accuracy (0.9771). Beats v2 across the board, despite v2 carrying three additional components.

2. **Highest cancer-detection rate**

   Catches 315 of 337 nodules (Recall = 0.9347) — misses only 22. In medical screening, recall is the clinically critical direction; v1 + TTA wins it cleanly.

3. **Fewest false alarms too**

   Only 13 false-positive predictions (Specificity = 0.9891), the lowest of all six configurations. So we don't trade precision for recall — we improve both simultaneously.

4. **Simplest possible deployment**

   Same trained model as v1 — no new parameters, no retraining cost. TTA is just 5 inference passes averaged. Easiest model to maintain, audit, and explain.

Alternatives — v2 if CBAM interpretability heatmaps are required for clinical UI · **v3** for edge / low-VRAM deployment (3× smaller).

What we delivered

1

Leakage-free LUNA16 classifier

Patient-level split across full LUNA16 (888 patients). Test = 1 528 patches across 176 unseen patients — audited, zero overlap.

2

Three models + four-component ablation

v1 (ResNet-50) · v2 (+CBAM, Focal, MixUp, TTA) · v3 (DenseNet-121+CBAM, …) + v1 with each component in isolation.

3

Final deployment: v1 + TTA

F1 = 0.9474 · Recall = 0.9347 · MCC = 0.9329 · AUC = 0.9934 — catches 315 of 337 nodules with only 13 false alarms.

4

Honest scientific story

Caught patient-level leakage on IQ-OTH/NCCD, caught a metric-reporting bug ourselves during sanity-check, reported corrected numbers with CIs.

Live Demo (≤ 1 minute)

60 s live inference · 6 unseen test patches · CBAM spatial-attention heatmap overlaid · *model = v2 (interpretability illustration); deployment model is v1 + TTA.*

Team Roles & What Each Member Did

Member	Student ID	Role	Contribution
Huu Thuc Tran	26164741	***Leader***	Dataset acquisition + extraction; patient-leakage analysis on IQ-OTH/NCCD; LUNA16 patch pipeline; implementation, training and evaluation of v1, v2, v3 (CBAM, Focal Loss, MixUp, TTA); 9-metric comparison study; final report.
Alex	26131779	*Member*	Slide design (Seminar 1) and initial Seminar 2 draft; early literature review on lung cancer detection.
Rabiul	*[Student ID — TBC]*	*Member*	Literature review on lung-nodule classification methods.

Team-roles statement: the Leader handled implementation and analytical work; teammates contributed to communication and literature support. *All members agreed the contribution split above is accurate.*

Thank you

Questions & Discussion